B.M.S. College of Engineering

Bio Inspired Algorithm

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Stock Price Prediction for Indian Banking Sector Using LSTM Neural Networks Optimized with Genetic Algorithm

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1 .Introduction to Optimization Problems

- What is Optimization?
  - Finding the best solution from all possible solutions
  - Minimizing or maximizing an objective function
  - Subject to constraints and limitations
- Types of Optimization Problems:
  - Continuous: Real-valued parameters (e.g., learning rate, dropout)
  - Discrete: Integer values (e.g., number of layers, neurons)
  - Mixed: Combination of both
  - Multi-objective: Multiple competing goals

2 .Nature Inspired Algorithms

- Definition:
  - Optimization algorithms inspired by natural phenomena
  - Mimic biological, physical, or social processes
  - Metaheuristic approaches for complex problems

- Popular Nature-Inspired Algorithms:
  - Genetic Algorithm (GA): Evolution and natural selection
  - Particle Swarm Optimization (PSO): Bird flocking behavior
  - Ant Colony Optimization (ACO): Ant foraging patterns
  - Simulated Annealing (SA): Metallurgical annealing process
  - Differential Evolution (DE): Population-based optimization

- Key Advantages:
  - No gradient information required
  - Escape local optima through randomization
  - Handle discrete, continuous, and mixed parameters
  - Parallel evaluation of multiple solutions
  - Robust to noisy fitness evaluations

3 .Problem Statement

- Research Challenge:
  - Stock market prediction is highly complex and non-linear
  - LSTM networks effective but performance depends on hyperparameters
  - Manual tuning is time-consuming, biased, and suboptimal

- Specific Problem:
  - Predict stock prices for 5 major Indian banks
  - Optimize LSTM hyperparameters automatically
  - Achieve superior accuracy compared to manual configuration

- Target Banks:
  - HDFC Bank (HDFCBANK. NS)
  - ICICI Bank (ICICIBANK.NS)
  - State Bank of India (SBIN. NS)
  - Axis Bank (AXISBANK.NS)
  - Kotak Mahindra Bank (KOTAKBANK.NS)

4 .DATASETS AND ITS PREPROCESSING

- Dataset:
  - Period: 8 years (2017-2025)
  - Frequency: Daily closing prices
  - Features: 35+ (Price, Volume, Technical Indicators)
  - Total Data Points: ~2,100 per stock

- Research Objectives:
  - Develop GA-based hyperparameter optimization framework
  - Design chromosome encoding for LSTM parameters
  - Achieve statistically significant performance improvement
  - Validate practical utility through trading strategies

5 .Implementation and Results

- System Architecture:
  - Data collection via Yahoo Finance API
  - Feature engineering with technical indicators
  - GA optimization framework
  - LSTM model training and evaluation
  - Performance analysis and visualization

- Hyperparameters Optimized (7 genes):
  - Number of LSTM layers: [1-3]
  - Neurons in layer 1: [32-256]
  - Neurons in layer 2: [32-256]
  - Learning rate: [0.0001-0.01]
  - Batch size: [16-128]
  - Dropout rate: [0. 1-0.5]
  - Lookback window: [30-120 days]

- GA Configuration:
  - Population size: 30 individuals
  - Generations: 20
  - Crossover rate: 80%
  - Mutation rate: 15%
  - Selection: Tournament (k=3)
  - Elitism: Top 2 survivors

6 .Performance Metrics

- Evaluation Metrics Used:
  - RMSE: Root Mean Square Error
  - MAE: Mean Absolute Error
  - R^2: Coefficient of Determination
  - MAPE: Mean Absolute Percentage Error

- Directional Accuracy: Trend prediction correctness

- Convergence Analysis:
  - 90% optimal fitness reached by generation 12-15
  - Population diversity: 0.85 → 0.32 (healthy convergence)
  - Computational efficiency: 10. 8 minutes per individual

7.Advantages

- Advantages:
  - Superior Accuracy: 23% improvement over manual tuning
  - Automation: No manual hyperparameter tuning required
  - Robustness: Consistent across multiple runs (low variance)
  - Flexibility: Handles mixed parameter types (continuous + discrete)
  - Practical Utility: 65. 9% directional accuracy enables trading
  - Multi-objective: Balances accuracy, complexity, training time
  - Scalability: Easily extends to other stocks and sectors
  - Interpretability: Feature importance analysis available

7.Limitations

- Limitations:
  - Computational Cost: 4-6 hours for complete optimization
  - Data Requirements: Needs 3-5 years historical data minimum
  - Black Box: Limited interpretability of LSTM predictions
  - Structural Breaks: Cannot anticipate unprecedented events
  - Market Regime Dependency: Lower accuracy during high volatility
  - Overfitting Risk: Requires careful validation
  - Retraining Needs: Quarterly updates recommended
  - No Causal Inference: Correlations, not causation

THANK YOU